

BHABHA UNIVERSITY, BHOPAL

NH-12, Hoshangabad Road, Jatkhedhi, Bhopal-462026

MASTER IN COMPUTER APPLICATION (MCA)

Forth Semester
Scheme of Examination



DETAILED SYLLABUS
(EFFECTIVE FROM JULY 2018)

Department of Computer Science & Application
BHABHA UNIVERSITY, BHOPAL

MASTER IN COMPUTER APPLICATION (MCA)

FOURTH SEMESTER **Scheme of Examination**

S.No	Subject Code	Subject Name	Periods per week			Credits	Maximum Marks (Theory slot)			Maximum Marks (Practical Slot)		Total Marks
			L	T	P		End Sem Exam	Tests	Assignment / Quiz	End Sem Practical Viva	Practical Record/Assignment/Quiz/Presentation	
1	MCA41	Computer Network	3	1	-	4	70	20	10		-	100
2	MCA42	Computer Graphics & Multimedia	3	1	-	4	70	20	10	-	-	100
3	MCA43	Artificial Intelligence	3	1	-	4	70	20	10	-	-	100
4	MCA44	Design and analysis of algorithm	3	1	-	4	70	20	10	-	-	100
5	MCA45	Elective I	3	1	-	4	70	20	10	-	-	100
6	MCA46	MINOR PROJECT-I	-	-	8	8	-	-	-	100	50	150
7	MCA47	Computer Graphics & Multimedia lab	-	-	2	2	-	-	-	30	20	50
		Total	15	5	10	30	350	100	50	130	70	700

L: Lecture - T: Tutorial - P: Practical

Elective I

- Mobile communication
- Compiler Design
- Dot Net Technology

MCA41- Computer Networks

UNIT-I

Use of communication and IT, Communication Mode- Simplex, Half Duplex, Full Duplex, Communication Channels - Twisted, Coaxial, Fiber Optic, Serial and Parallel Communication, Types of Network - LAN, WAN, MAN ,Internet etc., Topologies of LAN - Ring, Bus, Star, Mesh and Tree topologies, World Wide Web Internet Services, Analog & Digital Signal.

UNIT-II

Base Band , Broad Band, Multiplexer FDM, TDM, Modulation AM, FM, PM, Transmission Media ,Modem. OSI Reference Model , Switching Technique, Message Switching, Circuit Switching, Packet Switching, Virtual Circuit, , IEEE Standards, 802.3, 802.4, 802.5.

UNIT-III

Fast Ethernet, FDDI Token Ring, Wireless LAN, Inter-Networking Devices, Bridge, Routers Gateways, Repeater, Routing Algorithms, Distance Vector Routing, Shortest Path Routing, Broadcast Routing, Multicast Routing, TCP/IP Protocol, IPV4 Addressing, Congestion Control, Traffic Shaping.

UNIT-IV

Comparison between OSI and TCP/IP Models, TELNET, FTP, SMTP, MINE, UDP, URL (Uniform Resource Locator) HTTP, ISDN Channel, ISDN Services, Base Band ISDN, Broadband ISDN.

UNIT-V

Network Security : Network Security Issues, Firewalls – Need and Features of Firewalls, Types of Firewall Technology- Network Level and Application Level, IP Packets Filter Screening Routers, Limitations of Firewalls.

TEXT & REFERENCE BOOKS :

- DATA COMMUNICATION BY PRAKASH C GPTA
- COMPUTER NETWORKING BY ANDREWS TANANBAUM
- UNDERSTANDING DATA COMMUNICATION OF NETWORKING BY WILLIAM A SHAY
- COMMUNICATION AND NETWORK BY LEWIS MACHENZIE
- DATA AND COMPUTER COMMUNICATION: BY WILLIAM STALLINGS

MCA42- Computer Graphics & Multimedia

UNIT-I

Computer Graphics: Introduction , classification & Applications, Development of Hardware & Software for Computer Graphics. Display devices, Hard copy devices. Interactive Input devices, display processor, Line drawing; various algorithms and their comparison, circle generation- Bresenham's mid point circle drawing algorithm, mid point ellipse drawing algorithm.

UNIT-II

Attributes of output primitives, line style, color and intensity, Area filling algorithms, Scan line algorithm, and boundary fill flood fill algorithm. Two dimensional transformations; translation, scaling, rotation, reflection, composite transformation, transformation commands, character generation.

UNIT-III

Viewing coordinates, Window, view port, clipping, Window to view port transformation, line clipping algorithm; Cohen Sutherland, polygon clipping; Sutherland hodgman algorithm, 3D clipping : Normalized view volumes, view port clipping, clipping in homogeneous coordinates. Illumination model: Light sources, diffuse reflection specular reflection, reflected light, intensity levels, surface shading; phong shading ground shading, color models like RGB, YIQ, CMY, HSV etc.

UNIT-IV

3-D Viewing: Three-dimensional concepts, 3D display techniques, 3D representation polygon & curved surfaces. Design of curves & surfaces- Bezier's Method, B-spline methods, 3D transformation transition ,scaling, composite transformation rotation about arbitrary axis, projections: Parallel & Perspective, Hidden surface and line removal buffer and scan line methods.

UNIT-V

Introduction to multimedia, multimedia components, multimedia hardware, SCSI, IDE, MCI, Multimedia data and file formats, RTF, TIFF, MIDI, JPEG, DIB, MPEG, Multimedia tools, presentations tools, Authoring tools, presentations.

BOOKS

- 1 Roger S David "Mathematical Elements for Computer Graphics", McGraw Hill.
2. S. Harrington – "Computer Graphics - a Programming approach" (2nd ed) McGrawhill.
3. New Mann & Sprovl- "Principles of interactive computer graphics" (2nd ed) McGrawhill.
4. Roger S. David "Procedural Elements for Computer Graphics", McGraw Hill.
5. Foley & Vandan "Computer Graphics Principles & Practice in "C" "Addision Wesly.
6. Tay Vaughan " Multimedia Making it Work" 5th Ed. 2001, Tata McGraw Hill.

7. Prabhat K. Andleigh & Kiran Thakur “Multimedia System Design”, PHI
8. Drew, “Fundamentals of Multimedia”, Pearsons.

MCA43 - Artificial Intelligence

UNIT-I

General Issues and Overview of AI: The AI problems, what is an AI technique, Characteristics of AI applications. Introduction to LISP programming: Syntax and numeric functions, Basic list manipulation functions, predicates and conditionals, input output and local variables, iteration and recursion, property lists and arrays.

UNIT-II

Problem Solving, Search and Control Strategies: General problem solving, production systems, control strategies forward and backward chaining, exhaustive searches depth first breadth first search.

Heuristic Search Techniques: Hill climbing, branch and bound technique, best first search & A* algorithm, AND / OR graphs, problem reduction & AO* algorithm, constraint satisfaction problems.

UNIT-III

Knowledge Representations: First order predicate calculus, skolemization, resolution principle & unification, interface mechanisms, horn's clauses, semantic networks, frame systems and value inheritance, scripts, conceptual dependency.

UNIT-IV

Natural Language processing: Parsing techniques, context free grammar, recursive transition nets (RNT), augmented transition nets (ATN), case and logic grammars, syntactic analysis .Game playing Minimax search procedure, alpha-beta cutoffs, additional refinements.

Planning: Overview an example domain the block world, component of planning systems, goal stack planning, nonlinear planning.

UNIT-V

Probabilistic Reasoning and Uncertainty: Probability theory, bayes theorem and bayesian networks, certainty factor.

Expert Systems: Introduction to expert system and application of expert systems, various expert system shells, vidwan frame work, knowledge acquisition, case studies, MYCIN.

BOOKS

1. Elaine Rich and Kevin Knight “Artificial Intelligence” - Tata McGraw Hill.
2. “Artificial Intelligence” 4 ed. Pearson.
3. Dan W. Patterson “Introduction to Artificial Intelligence and Expert Systems”, Prentice India.
4. Nils J. Nilson “Principles of Artificial Intelligence”, Narosa Publishing House.
5. Clocksin & C.S.Melish “Programming in PROLOG”, Narosa Publishing House.

MCA44 - Design and analysis of algorithm

UNIT – I

Pre-requisites: Data structure & Discrete structures, models of computation, algorithm analysis, order architecture, time space complexities average and worst case analysis.

UNIT-II

Divide and conquer: Structure of divide-and-conquer algorithms: examples; Binary search, quick sort,

Strassen Multiplication; Analysis of divide and conquer run time recurrence relations.

Graph searching and Traversal: Overview, Traversal methods (depth first and breadth first search)

UNIT-III

Greedy Method: Overview of the greedy paradigm examples of exact optimization solution (minimum

cost spanning tree), Approximate solution (Knapsack problem), Single source shortest paths.

Branch and bound: LC searching Bounding, FIFO branch and bound, LC branch and bound application:

0/1 Knapsack problem, Traveling Salesman Problem, searching & sorting algorithms.

UNIT-IV

Dynamic programming: Overview, difference between dynamic programming and divide and conquer,

Applications: Shortest path in graph, Matrix multiplication, Traveling salesman Problem, longest Common sequence.

Back tracking: Overview, 8-queen problem, and Knapsack problem

UNIT-V

Computational Complexity: Complexity measures, Polynomial Vs non-polynomial time complexity;

NP-hard and NP-complete classes, examples. Combinational algorithms, string processing algorithm, Algebraic algorithms, set algorithms

BOOKS

1. Ullman "Analysis and Design of Algorithm" TMH
2. Goodman "Introduction to the Design & Analysis of Algorithms, TMH-2002.
3. Sara Basse, A. V. Gelder, "Computer Algorithms," Addison Wesley
4. T. H. Cormen, Leiserson, Rivest and Stein, "Introduction of Computer algorithm," PHI
5. E. Horowitz, S. Sahni, and S. Rajsekarani, "Fundamentals of Computer Algorithms," Galgoti

MCA45 - Elective I (A) Mobile communication

Unit-I

Overview of OSI Model: Significance of layered Model, PDUs, SDUs, IDUs, Higher layer Protocols. Switching and Components. Introduction, Applications, history, of wired & wireless Communication Systems. Radio Transmission: frequencies ,signal propagation, antenna, types of modulation, FHSS, DSSS. Multiple Access technology for Wireless Communication: FDMA, TDMA, CDMA Cellular System: Introduction, types.

Unit-II

Mobile Data Communication: Cellular Telephony, Structure, Fading, Small scale fading, Multipath Fading, Speech Coding, Error Coding and Correction, Hand off Management, Switching and authentication, MTSO interconnections, frequency hopping, frequency reuse. Circuit Switched Data Services & Packet Switched Data Services on Cellular Networks, Personal Communication Systems (PCS) Architecture, Digital Enhanced Cordless Telecommunications (DECT,) Personal Access Comm. System (PACS).

Unit-III

Digital Cellular Systems and Standards: GSM System overview, Architecture, GSM Protocol Model, GSM Mobility Management, SMS security aspects. Broadcast System overview. General Packet Service (GPRS) Architecture, GPRS Network, Interfaces and Procedures (2.5 G), 3G Mobile Services: UMTS and International Mobile Telecommunications (IMT-2000), W-CDMA and CDMA 2000, Quality of service in 3G.

Unit-IV

WLAN: Components and working of Wireless LAN, Transmission Media for WLAN, Infrastructure & Types of WLAN, IEEE 802.11 Standards, Protocols for WLAN, MACA, MACAW, Infrared technology. Wireless Application Protocol (WAP), Gateway, WAP protocols and WML.

Unit-V

Introduction to Bluetooth technology, Wireless in local loop (WLL) architecture, products, Satellite as a switch, Components of VSAT system, VSAT topologies, access schemes.

Books

1. Jochen Schiller "Mobile Communication", Pearson Education.
2. Raj Panadaya "Mobile and Personal Communication System & services
3. Lee "Mobile Cellular Telecom" 1995 Mc Graw Hill

MCA45 - Elective I(B) Compiler Design

Unit-I

Introduction to Compiling and one pass compiler : Compilers and translators, phases of compilers, Structure of a compiler, compiler writing tools, bootstrapping, overview of one pass compiler, Error handling.

Finite Automata & Lexical Analysis : Role of lexical analyser, specification of tokens, recognition of tokens, regular expression, finite automata, form regular expression to finite automata, DFA and NFA, implementation of lexical analyser, tools for lexical analyser, only introduction to LEX.

UNIT-II

Syntax Analysis & Parsing Techniques : Context free grammars, Phase tree, ambiguity of parse tree, bottom up parsing and top down parsing, shift reduce parsing, operator precedence parsing, elimination of left recursion, recursive descent parsing, predictive parser construction, Transition diagram.

UNIT-III

LR parsers, constructing SLR and canonical LR parsing tables, using ambiguous grammar, Introduction to YACC, LR(1) & LALR Parsers.

Syntax Directed Translation: Syntax directed translation scheme, construction of syntax trees, SDT with inherited and synthesized attributes, symbol tables.

UNIT-IV

Intermediate code generation : Intermediate languages, prefix notation, three address code, quadruples and triples, translation of assignment statements, boolean expression, procedural calls and iterative statements.

Run time Environment : Source language issues, storage organisation and allocation strategies, parameter passing, implementation of block structured languages.

UNIT-V

Error Detection and Recovery: Errors, sources of errors, Lexical & syntactic phase error, semantic errors panic mode error recovery & phrase level error recovery mechanisms.

Code Optimization: Optimization of basic blocks, loop optimization, global data flow analysis, loop invariant computations and other related optimization techniques.

Code Generation : Issues in design of code generation, simple code generation techniques.

BOOKS

1. Alfred V. Aho, Ravi Sethi and J.D. Ullman “Compilers- Principles, Techniques and tools” Addison Wesley. A
2. Alfred V. Aho and J.D. Ullman “Principles of Compiler Design” Narosa Publishing House.

3. Tremblay, "Theory and Practice of compiler writing", Mc Graw Hill.
4. Holuv, "Compiler Design in C", PHI.

Note: Paper is to be set unit wise with internal choice.

MCA45 - Elective I(C) Dot Net Technology

UNIT-I

Introduction: .NET Technology, Introduction to VB.NET, Software development and Visual Basic .NET, Visual Basic .NET and .NET frame.

UNIT-II

Visual Basic fundamentals: The Visual Basic .NET Development Environment, The element of VB.NET, VB.NET Operators, Software design, Conditional structure and control flow, Methods.

UNIT-III

Classes and Objects: Types, Structure and Enumeration, Classes, Interfaces, Exception handling and Classes, Collections, Arrays and other Data Structure.

UNIT-IV

Advance design concepts, Patterns, Roles and Relationships, Advanced Interface Patterns: Adapters and Delegates and Events Data Processing and I/O.

UNIT-V

Writing Software with Visual Basic .NET, Interfacing with the End User, Introduction to ASP.NET and C#.NET and their features.

BOOKS

1. Jeffrey R. Shapiro "The Complete Reference Visual Basic .NET" Tata Mcgraw Hill (2002 Edition).
2. Rox "Beginner and Professional Edition VB.NET" Tata Mcgraw Hill.
3. Steven Holzner "Visual Basic .NET Black Book" Wiley Dreamtech Publication.
4. Alex Homer, Dave Sussman "Professional ASP.NET1.1" Wiley Dreamtech
5. Bill Evzen,Bill Hollis "Professional VB.NET 2003" Wiley Dreamtech
6. Tony Gaddis "Starting Out VB.NET PROG.2nd Edition" Wiley Dreamtech
7. Chris Ullman, Kauffman "Beg. ASP.NET1.1 with VB.NET 2003" Wiley Dreamtech